

# Fin\_Cash



## Capstone Portfolio Template 2026-2027

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**Grade: 12**

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## ABSTRACT

**Financials Hub** is a web-based system designed to **address the challenges schools face in managing financial records** efficiently. Many educational institutions still rely on spreadsheets and manual processes, which makes tracking transactions, managing approvals, and maintaining transparency difficult.

The system allows financial staff to create structured records and enter settlement data based on selected categories, while attaching supporting documents for each transaction. School principals can review, approve, or reject records, and communicate with staff directly within each record. School owners and board members can monitor financial activity through dashboards and summarized views.

By centralizing financial workflows and introducing clear tracking and role-based interaction, Financials Hub **helps reduce errors, improve transparency, and support faster, more informed decision-making within schools.**

## Introduction

Many schools still rely on spreadsheets, paper records, and manual processes to manage expenses and custody balances. Financial data is often scattered across different files or physical documents, making it difficult to maintain consistency and accuracy. In many cases, approval processes depend on handwritten signatures or informal communication methods, which slows down operations and increases the risk of miscommunication.

These limitations make it challenging to track the status of financial records, monitor ongoing transactions, and ensure accountability. As a result, the entire process becomes time-consuming, less transparent, and more prone to human error, which can negatively affect decision-making and overall financial control.

To address these challenges, our project, **Financials Hub**, provides a web-based system for managing cash disbursement records in a more structured and efficient way. The system allows financial staff to create organized records and enter categorized **settlement data**, while attaching supporting documents for verification. School principals can review, approve, or reject records through a clear workflow, and all actions are tracked within the system.

Additionally, the platform provides dashboards and summarized views that help administrators and stakeholders monitor financial activity in real time. By centralizing financial processes and improving communication between users, Financials Hub **enhances transparency, reduces errors, and enables faster and more reliable financial management within schools.**

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## Team Roles and Responsibilities

Our team consists of the following members:

Marwan Hazem – Backend Developer

Responsible for developing and implementing the backend logic of the system,

including handling report processing, role-based permissions, and system functionality.

**Yomna Reda** – Backend Developer

Collaborated on backend development, ensuring proper integration between the database and the application logic.

**Sama Adel**– Frontend Developer

Responsible for developing the user interface using React and implementing interactive data tables using AG Grid.

**Youssef Mohamed** – Database Designer & Documentation

Responsible for designing the database structure, developing the system prototype using Figma, and preparing project documentation.

**Mohammed Waleed** – Tester & Project Coordinator

Responsible for testing system functionality, tracking team progress using the SCRUM methodology through Jira, and contributing to documentation.

Although each team member has a primary responsibility, collaboration is an essential part of our workflow. Team members support one another when tasks are completed or when assistance is needed. This flexible approach ensures balanced workload distribution and allows each member to gain experience in different aspects of the project.

## Table of Contents

|                                                         |   |
|---------------------------------------------------------|---|
| Section I: The problem background.....                  | 4 |
| 1.1 Sustainable Development Goals (SDGs) addressed..... | 5 |

|                                                  |    |
|--------------------------------------------------|----|
| 1.2 Empathize.....                               | 6  |
| Section II: Ideas and solutions.....             | 7  |
| 2.1 Define.....                                  | 8  |
| 2.2 Task Review:.....                            | 8  |
| 2.3 Problem Statement:.....                      | 8  |
| 2.4 Prior Solutions:.....                        | 8  |
| 2.5 Ideate:.....                                 | 10 |
| 2.6 Problem Solution:.....                       | 10 |
| 2.7 Our Selected Solution:.....                  | 10 |
| 2.8 Solution Advantages & Disadvantages:.....    | 10 |
| Section III: Experiment ideas and solutions..... | 12 |
| 3.1 Task Review:.....                            | 13 |
| 3.2 Methods used in Requirement Gathering:.....  | 13 |
| 3.3 Use Case Diagram:.....                       | 13 |
| 3.4 Activity Diagram:.....                       | 14 |
| 3.5 Sketches:.....                               | 15 |
| 3.6 Linkable Prototype.....                      | 15 |
| 3.7 Test and Evaluate:.....                      | 16 |
| 3.8 Refine the design.....                       | 19 |
| Section IV: Analyzing and communicating.....     | 20 |
| 4.1 Recommendations.....                         | 21 |
| 4.2 Learning Outcomes.....                       | 22 |
| Competency Related:.....                         | 22 |
| 4.3 Communication.....                           | 23 |
| 4.4 References.....                              | 24 |
| Section V: Business Model Canvas Template.....   | 25 |

## Section I: The problem background.

## 1.1 Sustainable Development Goals (SDGs) addressed

In this chapter, we will discuss Sustainable Development Goals that our project supports and how this project can help in sustainable development.



•SDG's addressed:

### 1. SDG 4: Quality Education

Financials Hub supports Quality Education by strengthening the financial management systems within schools. By providing a centralized and structured platform for financial recording, custody tracking, and approvals, the system **enables schools to allocate resources more efficiently** and operate in a more organized manner. Improved financial management **contributes to a more stable and effective educational environment**.

### 2. SDG 16: Peace, Justice, and Strong Institutions

The project aligns with SDG 16 by promoting transparency, accountability, and structured governance within school institutions. Through role-based access, approval workflows, and real-time tracking of financial records, Financials Hub **enhances institutional oversight and reduces the risk of financial mismanagement**. This **supports the development of effective and accountable institutions**.

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### 3. SDG 9: Industry, Innovation, and Infrastructure

Financials Hub contributes to SDG 9 by **replacing outdated manual financial processes with a modern web-based system**. By integrating dashboards, automated notifications, and centralized reporting, the project represents a technological innovation that improves operational infrastructure within educational institutions.

## 1.2 Empathize

- **Target Users**

-  **Financial Staff (Accountants / Employees):**  
They are responsible for recording financial transactions and creating cash disbursement records. They enter transaction details, attach supporting documents, and manage settlement data within the system.
  -  **School Principals (Administrators):**  
They review submitted records, approve or reject them, and ensure that all financial activities follow the correct procedures.
  -  **School Owners / Board Members:**  
They have a monitoring role, where they review financial activity through dashboards and reports without directly interacting with the data entry process.
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- **User Needs and Challenges**

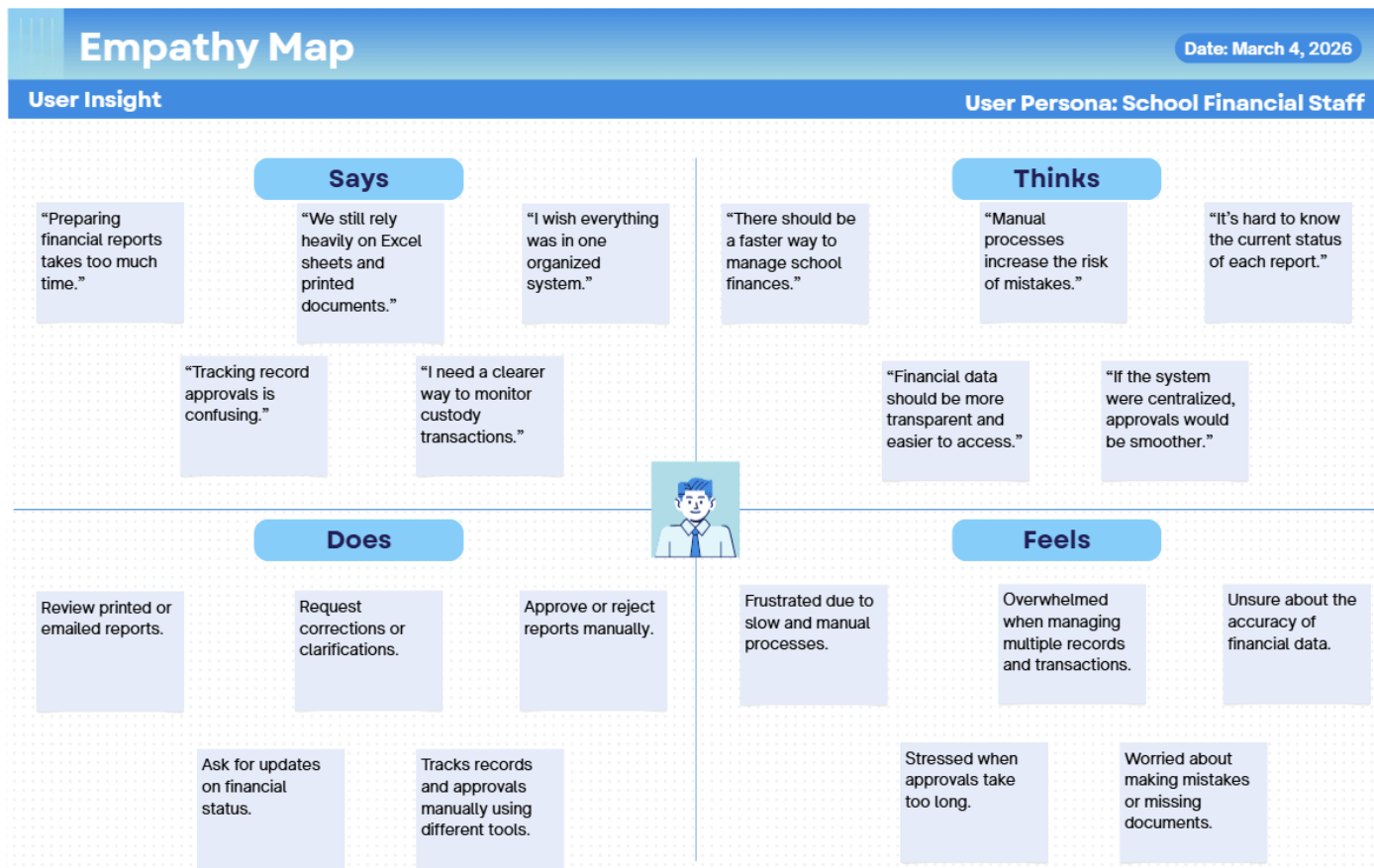
Our system is designed to support school financial operations by addressing the challenges faced by different user roles.

-  **Financial Staff (Accountants / Employees):**  
Staff often rely on spreadsheets, paper records, or messaging

applications to manage financial data. This can lead to confusion, errors, and difficulty tracking transactions. They need a clear and organized way to create **cash disbursement records**, enter **settlement data**, and attach supporting documents. They also need a structured process that guides them through what actions are allowed at each stage. Financials Hub provides a centralized system that simplifies this process and reduces mistakes.

-  **School Principals (Administrators):**  
Principals may struggle to review financial records efficiently when data is not well organized. They need a clear view of all submitted **records**, along with the ability to approve, reject, or request changes easily. They also need a reliable way to communicate with staff and ensure that all actions are tracked. The system supports this by providing a structured workflow with status tracking and in-app communication.
-  **School Owners / Board Members:**  
Owners and board members need to stay informed about financial activity without being involved in daily operations. They need easy access to summaries and a clear overview of **cash disbursements**. Financials Hub provides dashboards and read-only access, allowing them to monitor performance and make informed decisions.

## Empathy Map



## Section II: Ideas and solutions

## 2.1 Define

### Final Problem Statement

Schools face challenges in managing financial operations due to their reliance on spreadsheets, paper records, and scattered communication methods. Financial staff struggle to organize cash disbursement records and track settlement data efficiently, while principals find it difficult to review and monitor records in a clear and timely manner. This results in delays, errors, and limited transparency across the financial workflow.

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### Impact on Users and the Market

Solving this problem will improve how financial operations are managed within schools. Financial staff will be able to create and manage records in a structured and organized way, reducing confusion and minimizing errors. Principals will gain better visibility and control over approvals, allowing them to make faster and more informed decisions. School owners and board members will benefit from clear dashboards that provide an overview of financial activity without being involved in daily operations.

On a broader level, this solution supports the digital transformation of educational institutions. By replacing manual processes with a centralized system, Financials Hub improves financial transparency, accountability, and efficiency. This helps schools operate more professionally and prepares them to adopt more advanced digital systems in the future.

## 2.2 Task Review:

In this chapter, we will discuss the problem definition, review prior solutions, and present the selected solution for our project.

Several possible solutions were considered to address the problem, including improving spreadsheet-based methods, using existing communication tools, and developing a dedicated digital system. Existing solutions such as spreadsheets and general accounting tools were also reviewed, but they lack structure and efficiency for school workflows.

Based on this, a web-based system was selected as it provides a more organized and effective approach. The prototype is designed to reflect this workflow through a

simple interface, and it will be tested for functionality and usability. The required materials are mainly digital tools, making the solution cost-effective.

## 2.3 Problem Statement:

The core problem our project aims to solve is the inefficiency and lack of organization in managing financial operations within schools. Many schools still rely on spreadsheets, paper records, and scattered communication methods, which makes it difficult to track financial records, monitor their status, and maintain clear communication between staff and administrators. This often leads to delays, errors, and limited transparency.

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### Impact on Users and the Market

Solving this problem will significantly improve the way financial operations are handled in schools. Financial staff will be able to manage records more efficiently, reducing confusion and minimizing errors. Principals will gain better visibility over financial activities, allowing them to review and approve records more quickly and accurately. School owners and board members will also benefit from clear insights into financial performance.

On a broader level, this solution supports the transition toward digital systems in educational institutions. By improving organization and transparency, it helps schools operate more professionally and encourages the adoption of modern financial management tools.

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### How the Solution Improves User Experience and Industry

Financials Hub will make financial management easier and more reliable by replacing manual processes with a centralized system. Users will save time, reduce errors, and experience less stress when handling financial data. The system also improves communication and ensures that all actions are clearly tracked.

From an industry perspective, the solution contributes to the digital transformation of schools and promotes better financial practices. It can also create opportunities for further development of similar systems in educational institutions, supporting growth in the field of digital financial management.

## 2.4 Similar Solutions:

Several existing systems provide financial management features for schools. However, most of them are designed as broad, all-in-one platforms rather than focusing specifically on simple financial workflows.

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### Solution 1: Classter

A comprehensive school management system that integrates financial tools such as billing, budgeting, and reporting within a larger platform.

| Advantages                                 | Disadvantages                                        |
|--------------------------------------------|------------------------------------------------------|
| Offers a wide range of integrated features | Complex interface and steep learning curve           |
| High level of customization                | Onboarding process can be challenging                |
| Centralized data management                | Financial reports may require additional adjustments |

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## Solution 2: Schoology Learning

A learning management system that includes basic financial tracking features alongside its educational tools.

| Advantages                                | Disadvantages                                     |
|-------------------------------------------|---------------------------------------------------|
| User-friendly interface                   | Financial features are limited in depth           |
| Supports real-time tracking of data       | Not specifically designed for financial workflows |
| Helps in organizing resources efficiently | Some features can be difficult to navigate        |



### Solution 3: Gradelink SIS

A student information system designed for K–12 schools, offering tools for billing, invoicing, and financial tracking.

| Advantages                              | Disadvantages                                        |
|-----------------------------------------|------------------------------------------------------|
| Tailored for school environments        | Interface may feel outdated                          |
| Simplifies billing and payment tracking | Limited flexibility in some features                 |
| Improves accuracy of financial records  | Not fully focused on streamlined financial workflows |

## 2.5 Ideate:

To address the challenges in managing financial operations in schools, several possible solutions were explored. Each idea was evaluated based on simplicity, efficiency, and suitability for school environments.

| Possible Solutions            |                                                                               |                             |                                                 |
|-------------------------------|-------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------|
| Solution                      | Description                                                                   | Advantages                  | Disadvantages                                   |
| Spreadsheet-Based Improvement | Enhancing existing spreadsheet systems with better templates and organization | Easy to implement, low cost | Still manual, prone to errors, limited tracking |
| Using Communication           | Managing                                                                      | Familiar tools,             | No structure, hard to                           |

| Possible Solutions                         |                                                                                     |                                                          |                                                         |
|--------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|
| Tools                                      | financial records through messaging or email platforms                              | easy communication                                       | track records, inefficient                              |
| Full School Management System              | Using existing all-in-one platforms for financial management                        | Comprehensive features, centralized system               | Complex, overwhelming, not focused on finance workflows |
| Dedicated Web-Based System (Selected Idea) | A system specifically designed to manage financial records and workflows in schools | Organized workflow, role-based access, improved tracking | Requires development and initial setup                  |

## 2.6 Problem Solution:

The proposed solution is Financials Hub, a dedicated web-based financial management system tailored for educational institutions. Unlike general accounting software, Financials Hub focuses on the specific workflow of school cash disbursements and settlements.

The system provides a structured environment where Financial Staff can record transactions, categorize expenses (e.g., maintenance, supplies, activities), and upload digital evidence (invoices/receipts). This data is then routed through an automated Approval Workflow, allowing Principals to review and take action (Approve/Reject/Comment) in real-time. By centralizing these operations, the system eliminates the need for physical paperwork and fragmented communication, ensuring that every financial action is logged, transparent, and easily auditable.

## 2.7 Our Selected Solution:

Our selected solution is a Custom Web-Based Application built with a robust backend and an interactive frontend.

### Key Features of the Selected Solution:

- Role-Based Access Control (RBAC): Distinct interfaces and permissions for Staff, Principals, and Board Members.
- Automated Settlement Workflow: A step-by-step process for creating, submitting, and approving cash disbursement records.
- Digital Evidence Management: Integrated file uploading for attaching invoices directly to records.
- Real-Time Dashboard: Visual summaries of record statuses (Pending, Approved, Rejected) to support administrative decision-making.
- In-App Communication: A commenting system within each record to allow direct feedback between staff and principals.

## 2.8 Solution Advantages & Disadvantages:

| Advantages                                                                                   | Disadvantages                                                                                  |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Centralization: All financial data and documents are stored in one secure location.          | Initial Setup: Requires time to input initial school data and user roles.                      |
| Transparency: Real-time tracking of approval statuses and clear audit trails.                | Internet Dependency: Requires a stable internet connection to access the web platform.         |
| Efficiency: Reduces the time spent on manual data entry and physical document handling.      | Learning Curve: Staff accustomed to paper may need brief training on the new digital workflow. |
| Accuracy: Minimizes human error through structured forms and automated balance calculations. |                                                                                                |

## 2.9 Prototype Design:

The Financials Hub prototype is a web-based application accessible through a standard browser, built with a React.js frontend and an ASP.NET Core (C#) backend. It is designed to demonstrate the core financial workflow of a school — from cash

disbursement record creation by an accountant to review and approval by a principal — within a clean, role-sensitive interface.

## Core Pages and Features

**Login and Sign-Up Pages:** A pair of styled entry screens where users authenticate using their credentials. Upon login, the system identifies the user's role and routes them to the appropriate interface and permission set.

**Dashboard:** The landing page after login. It provides an overview of financial activity through summary cards (total records, pending approvals, approved/rejected counts) and a visual chart showing the distribution of expenses by category. The dashboard also features a search bar with date-range and category filters, a reset button, and a data-refresh button.

**Cash Disbursement Records Page (سجل صرف النقدية):** A table-based page listing all cash disbursement records available to the logged-in user. Each row displays the record's date, category, total amount, and current status (Draft, Under Review, Approved, Rejected, Deleted). The page includes an 'Add Cash Disbursement' button visible to accountants and a checkbox beside each record for bulk actions.

**Settlement Data Page (بيانات التسوية):** The main record creation and editing screen. Its title updates dynamically to reflect the selected category — for example, 'بيانات تسوية اشتراكات نت' when the Internet Subscriptions category is selected. The page contains a transaction table (built with AG Grid) where the accountant enters individual expense rows, each with a date, amount, description, and optional file attachment. Below the transaction table, an automatically calculated summary section displays the total amount paid, the previous balance, and the current balance. A comments and feedback section at the bottom enables direct communication between the accountant and the principal within the record.

**Notifications Page:** A dedicated page accessible to all users that lists in-app notifications triggered by status changes and messages. This replaced the profile page following testing feedback.

**Role-Based Navigation:** The sidebar navigation adapts based on the logged-in user's role. Accountants see options to create and manage records. Principals see a view focused on records awaiting review. Board Members have read-only access to the dashboard and financial data.

## 2.10 Prototype Testing:

To evaluate the functionality and usability of the Financials Hub prototype, we conducted structured testing across all core system features. Testing was performed by the team's designated tester, with other members serving as test users to simulate realistic usage across different roles.

## What Was Tested

- Authentication: login and sign-up page functionality and role-based routing.
- Dashboard: search filters by category and date range, the reset button, and the data-refresh button.
- Cash Disbursement Records Page: filtering by category and status, the reset filter button, and record visibility per role.
- Settlement Data Page: record creation, expense row entry, attachment upload and access, automatic summary calculation accuracy, and dynamic page title update based on selected category.
- In-App Chat and Notifications: message delivery between the accountant and principal roles, and notification generation upon status changes and new messages.
- Status Change Workflow: transitioning a record from Draft to Under Review, and from Under Review to Approved or Rejected by the principal.

## How Testing Was Conducted

Testing was performed by running each identified use case manually, with pre-defined input data and clearly documented expected results. For each test case, the tester recorded the actual result and compared it against the expectation. Any deviation was logged as either 'Not Pass' or 'Page not found' and added to the refinement list.

Role-based testing required the tester to log in separately as each user type — Accountant, Principal, and Board Member — and confirm that the visible pages, available actions, and data access matched the intended role configuration.

Attachment testing required creating a dedicated uploads folder in the backend hosting environment (MonsterASP) to resolve the 'Page not found' error that occurred when clicking on uploaded files.

## What We Evaluated

- Functionality: Do all system features behave as expected under normal conditions?
- Role Accuracy: Does the system correctly restrict or grant access based on the logged-in user's role?

- Data Integrity: Does the automatic summary in the Settlement Data page reflect the correct totals based on entered transaction rows?
- Communication Flow: Do messages sent within a record trigger the correct notifications for the receiving user?
- Filter Accuracy: Do the search and filter tools return only the records that match the selected criteria?

## 2.11 Materials and Costs:

Since Financials Hub is a software-based system, all tools required to build and run the prototype are digital. The following table lists each tool or resource used, its purpose, and its associated cost. All tools were either free, open-source, or available under student or free-tier licensing, making the overall development cost of the prototype effectively zero.

| Tool/Resource                    | Purpose                                                | Cost                |
|----------------------------------|--------------------------------------------------------|---------------------|
| Visual Studio Code (VS Code)     | Code editor for front-end and back-end development     | Free                |
| React.js                         | Front-end framework for building the user interface    | Free (open source)  |
| AG Grid (Community Edition)      | Interactive data table component used in records pages | Free                |
| C#/ <a href="#">ASP.NET</a> Core | Back-end framework for API and business logic          | Free (open source)  |
| SQL Server (Express Edition)     | Database management and storage                        | Free                |
| Figma                            | UI wireframe and prototype design                      | Free (student plan) |
| Git/GitHub                       | Version control and team collaboration                 | Free                |

|                               |                                                  |                      |
|-------------------------------|--------------------------------------------------|----------------------|
| MonsterASP (Back-end Hosting) | Deploying the back-end API for live testing      | Free tier available  |
| Jira (Student License)        | SCRUM-based project tracking and sprint planning | Free (via Atlassian) |

The only non-digital cost was time — the hours invested by each team member across development, testing, and documentation. No physical materials were required. This makes Financials Hub a genuinely low-cost prototype that could be scaled or commercialized with a relatively modest initial investment in cloud hosting and a paid hosting plan.

## 2.12 Preliminary Feedback

### Feedback Received from the Panel

During the preliminary review of our project, the supervising panel provided the following feedback:

- The project terminology needed to be made more realistic. The panel noted that describing the system as a 'Financial Report' system was too generic and did not reflect an actual school financial workflow. They directed us to reframe the system around the concept of 'Cash Disbursements' and 'Settlement Data,' which are terms that align more closely with how school finances are actually managed.
- The dashboard was considered incomplete without visual data representation. The panel specifically requested that charts be added to the dashboard — such as a pie chart or bar chart — to give principals and board members a meaningful visual summary of financial activity, rather than displaying only text-based information.
- The navigation structure required simplification. The panel observed that some of the earlier page designs, including a popup window for creating new records and a profile page accessible from the sidebar, were adding unnecessary steps to the core workflow. They advised us to streamline navigation and prioritize the most frequently used pages.
- The role-based logic had to be more clearly enforced. The panel emphasized that each user role — Accountant, Principal, and Board Member — should have a clearly distinct experience, with visible differences in what actions are available and what information is displayed.

### Changes Made Based on Panel Feedback

In response to the panel's feedback, we refined both the system's terminology and user experience to better align with real-world financial workflows in schools. The following improvements were implemented:

- **Terminology Alignment:**

All project terminology was revised to match the panel's direction. "Financial Report" was replaced with "**Cash Disbursement**," "Create Report" became "**Add Cash Disbursement**," and the data entry page was renamed "**Settlement Data**."

Additionally, the page title now updates dynamically based on the selected expense category, improving clarity for users.

- **Dashboard Enhancements:**

A pie chart was added to the dashboard to visualize the distribution of expenses across categories. This allows principals and board members to quickly understand spending patterns without opening individual records.

- **Improved Data Entry Experience:**

The popup window for record creation was removed and replaced with a full-page **Settlement Data** screen. This provides more working space and makes the process more structured and user-friendly.

- **Navigation Optimization:**

The profile page was removed from the sidebar and replaced with a **Notifications** page, which better supports the system's communication and approval workflow.

- **Role-Based Access Improvements:**

The role-based interface logic was strengthened so that each user type (Accountant, Principal, Board Member) sees a tailored version of the system. Access control is now enforced at both the frontend routing level and the backend API layer.

**Overall, these changes improved clarity, usability, and alignment with real institutional workflows.**

## Section III: Experiment ideas and solutions

### 3.1 Task Review:

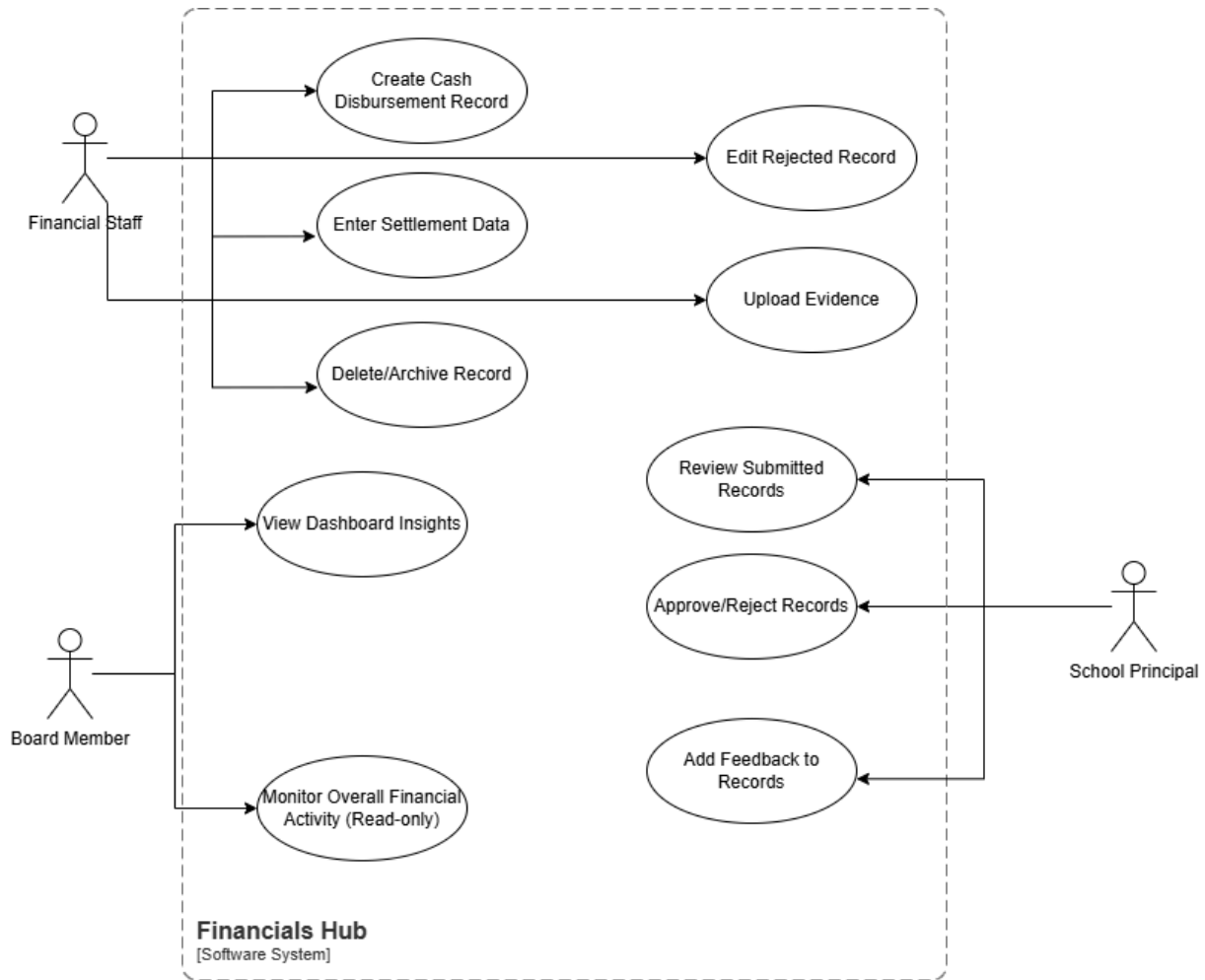
In this section, we transition from conceptual ideas to technical design. We gathered requirements through stakeholder interviews and translated them into system diagrams and prototypes. These models ensure that the system logic aligns with the actual needs of a school's financial department.

### 3.2 Methods used in Requirement Gathering:

- Stakeholder Interviews: Conducted discussions with school financial administrators to understand the current "pain points" in their manual processes.
- Document Analysis: Reviewed existing paper-based "Cash Disbursement" forms and Excel templates to identify necessary data fields.
- Observation: Observed the physical workflow of document submission and approval to replicate it digitally.

### 3.3 Use Case Diagram:

The Use Case Diagram for Financials Hub defines the functional requirements of the system by illustrating the interactions between external actors and the internal system processes. The diagram highlights a sophisticated, role-based architecture that prioritizes transparency and real-time data access.



## Use Case Description:

### System Actors & Shared Functionalities

The system identifies three primary actors: the Financial Staff (Accountant), the School Principal, and the Board Member. To ensure maximum institutional efficiency, the system features key shared use cases:

- **Receive Notifications:** All actors are alerted in real-time to status changes (submissions, approvals, rejections), ensuring a responsive workflow.
- **View Dashboard Insights:** This is a centralized functionality accessible to all roles. While the Board Member uses it for high-level oversight, the Principal utilizes it to monitor approval queues, and the Accountant tracks submission trends and pending records. This shared access ensures that all stakeholders are aligned with the institution's financial status.

### Core Use Cases by Role

#### 1. Financial Staff (Accountant) Operations

The Accountant manages the foundational data of the system:

- Create Cash Disbursement Record: Initiates new financial transactions.
- Enter Settlement Data & Upload Evidence: Ensures records are complete and auditable.
- Edit Rejected Record: Facilitates the feedback loop by refining records based on Principal comments.
- Delete/Archive Record: Manages the lifecycle of draft or completed entries.

## 2. School Principal Operations

The Principal provides administrative validation:

- Review Submitted Records: Evaluates "Under Review" entries for accuracy.
- Approve/Reject Records: Executes final decisions on disbursement requests.
- Add Feedback to Records: Provides professional guidance to the staff within the system.

## 3. Board Member Operations

The Board Member maintains strategic oversight:

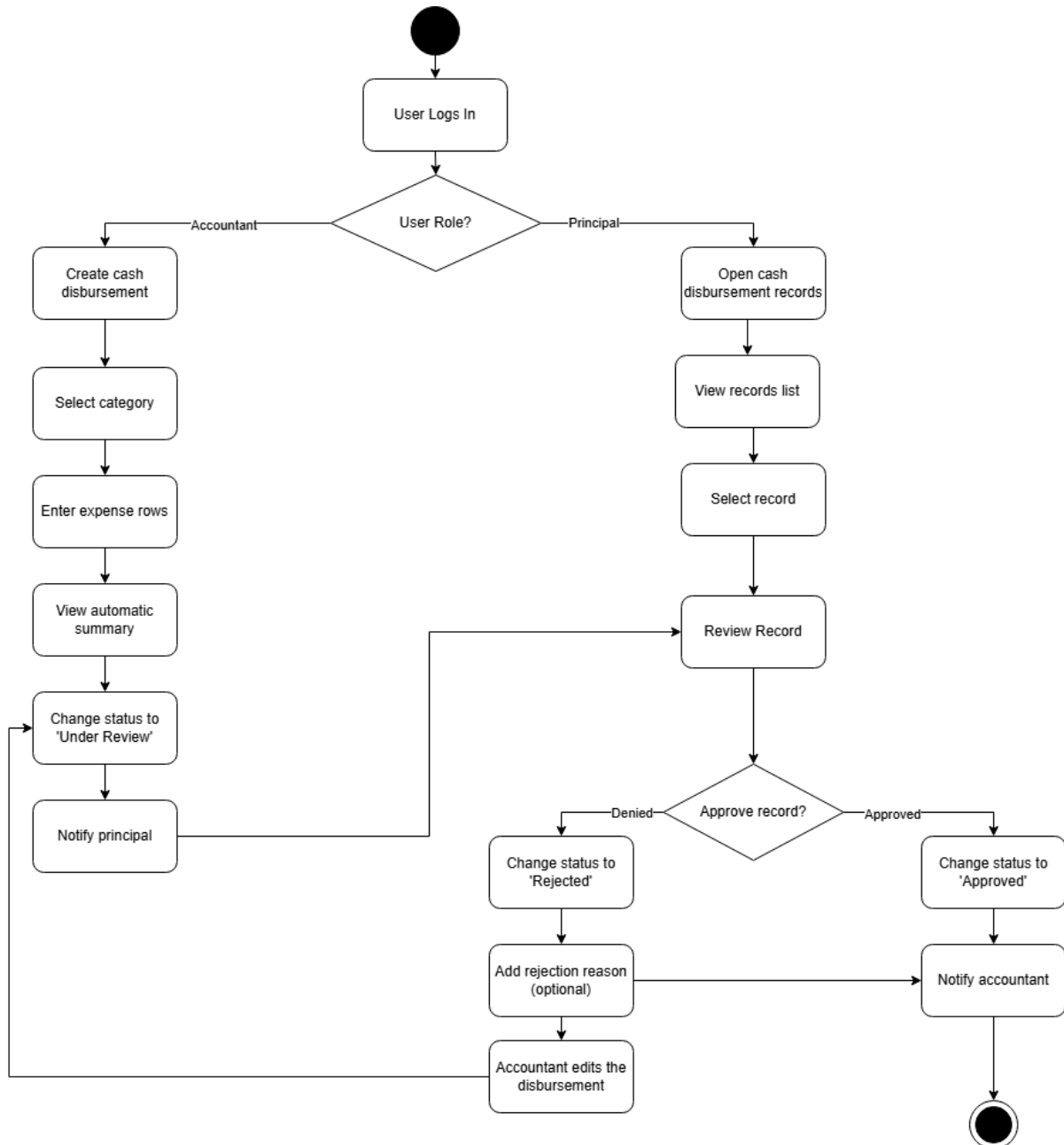
- Monitor Overall Financial Activity (Read-only): Provides a comprehensive, unalterable view of all institutional financial records to ensure total transparency.

## Conclusion

By integrating shared functionalities like automated notifications and a universal dashboard, Financials Hub creates a collaborative environment that eliminates information silos. This design fulfills the "Advanced" rubric criteria by demonstrating a well-reasoned, professional approach to school financial management.

### 3.4 Activity Diagram:

The Activity Diagram for **Financials Hub** illustrates the dynamic flow of the system's core financial workflow, specifically the "Cash Disbursement" process. It details the step-by-step interactions between the **Accountant** and the **Principal**, ensuring a transparent and auditable path from record creation to final approval.



#### Activity Diagram Description:

##### Process Overview

The workflow begins with a centralized Login action. Upon successful authentication, the system performs a Role Check to direct the user to their specific interface and set of permissions.

### 1. Accountant Workflow (Creation & Submission)

If the user is identified as an Accountant, they initiate the disbursement process:

- Create Cash Disbursement: The accountant starts a new record entry.
- Select Category: They categorize the expense (e.g., Maintenance, Supplies) to ensure organized financial tracking.
- Enter Expense Rows: Detailed transaction data is entered into the system.
- View Automatic Summary: The system automatically calculates and displays a summary of the entered data for verification.
- Change Status to 'Under Review': Once verified, the accountant submits the record, which automatically updates its status to "Under Review" (Pending).
- Notify Principal: The system triggers a notification to the Principal, signaling that a new record is ready for evaluation.

### 2. Principal Workflow (Review & Decision)

If the user is identified as a Principal, they manage the approval cycle:

- Open Cash Disbursement Records: The principal accesses the centralized repository of financial records.
- View Records List & Select Record: They browse the "Under Review" records and select a specific entry for detailed inspection.
- Review Record: The principal examines the data and attached evidence.
- Decision Point (Approve Record?):
- If Approved: The status is changed to "Approved," the record is finalized, and the accountant is notified of the success.
- If Denied (Rejected): The status is changed to "Rejected." The principal has the option to Add a Rejection Reason to provide feedback.
- Accountant Revision: In the case of rejection, the accountant is notified, edits the disbursement based on the feedback, and resubmits it, restarting the "Under Review" cycle.

### Conclusion

The process concludes once a record reaches the "Approved" state and the final notification is sent. This structured activity flow ensures that no financial record is finalized without proper oversight, fulfilling the institutional requirements for accountability and transparency.

### 3.5 Sketches:

تصدير بصيغة اكسل

+ إضافة تقرير

الموظف

سجل التقارير المالية

بحث في التقارير...

| التصنيف        | إجمالي المبلغ | الحالة       |
|----------------|---------------|--------------|
| إشتراكات نت    | £4,750.00     | معتمد        |
| مشتريات متنوعة | £12,500.00    | قيد المراجعة |
| عهدة           | £25,000.00    | مرفوض        |
| انتقالات       | £3,200.00     | مكتمل        |
| مشتريات متنوعة | £980.00       | مسودة        |
| إشتراكات نت    | £1,500.00     | محذوف        |

الإشعارات

إدارة جميع الإشعارات والتنبيهات

الكل (5)

غير مقروء (3)

مقروء (5)

تقرير جديد تم إنشاؤه

تم إنشاء تقرير 'بيانات تسوية عهدة مبلغ' بنجاح منذ 5 دقائق

مصرفات تنتظر الموافقة

هناك 3 مصرفات جديدة بحاجة إلى مراجعة وموافقة منذ 15 دقيقة

تحديث النظام

تم تحديث النظام إلى الإصدار الأخير بنجاح منذ ساعة

تقرير تمت الموافقة عليه

تمت الموافقة على تقرير 'مشتريات مكتبية متنوعة' منذ ساعتين

تنبيه: تقرير مرفوض

تم رفض تقرير 'عهدة نقدية ربع سنوية' - يرجى المراجعة

تصدير بصيغة اكسل + إضافة تقرير الموظف سجل التقارير المالية

بحث في التقارير...

| الحالة       | إجمالي المبلغ | التصنيف        |
|--------------|---------------|----------------|
| متمم         | £4,750.00     | إشتراكات نت    |
| قيد المراجعة | £12,500.00    | مشتريات متنوعة |
| مرفوض        | £25,000.00    | عهدة           |
| مكتمل        | £3,200.00     | انتقالات       |
| مسودة        | £980.00       | مشتريات متنوعة |
| محذوف        | £1,500.00     | إشتراكات نت    |

العودة إلى سجل التقارير

يمكنك إرفاق الملفات فقط - سيتم التوقيع التوقيع عند الإرفاق

بيانات تسوية إشتراكات نت

أضف وصفاً مختصراً (اختياري)

إشراكات نت التصنيف

أنشئ بواسطة أحمد يحيى

سجل المصروفات

إرفاق الملفات فقط

| م | التاريخ    | المبلغ | البيان                     | المرفقات            |
|---|------------|--------|----------------------------|---------------------|
| 1 | 01/20/2024 | 300    | شحن نت المدرسة 01252652642 | فاتورة.pdf<br>إرفاق |
| 2 | 01/22/2024 | 0.0    | أدخل البيان...             | إرفاق               |

## Final Materials List & Costs

Since Financials Hub is a software-based system, all tools used for development are digital. The table below lists the key resources required to build and host the prototype.

| Tool/Resource | Purpose | Cost |
|---------------|---------|------|
|---------------|---------|------|

|                                  |                                                        |                      |
|----------------------------------|--------------------------------------------------------|----------------------|
| Visual Studio Code (VS Code)     | Code editor for front-end and back-end development     | Free                 |
| React.js                         | Front-end framework for building the user interface    | Free (open source)   |
| AG Grid (Community Edition)      | Interactive data table component used in records pages | Free                 |
| C#/ <a href="#">ASP.NET</a> Core | Back-end framework for API and business logic          | Free (open source)   |
| SQL Server (Express Edition)     | Database management and storage                        | Free                 |
| Figma                            | UI wireframe and prototype design                      | Free (student plan)  |
| Git/GitHub                       | Version control and team collaboration                 | Free                 |
| MonsterASP (Back-end Hosting)    | Deploying the back-end API for live testing            | Free tier available  |
| Jira (Student License)           | SCRUM-based project tracking and sprint planning       | Free (via Atlassian) |

All tools used in this project were either free or available through student licenses. This makes the solution cost-effective and accessible for a school-level capstone project. The primary investment was time and effort from the development team.

### Expected Challenges During Testing

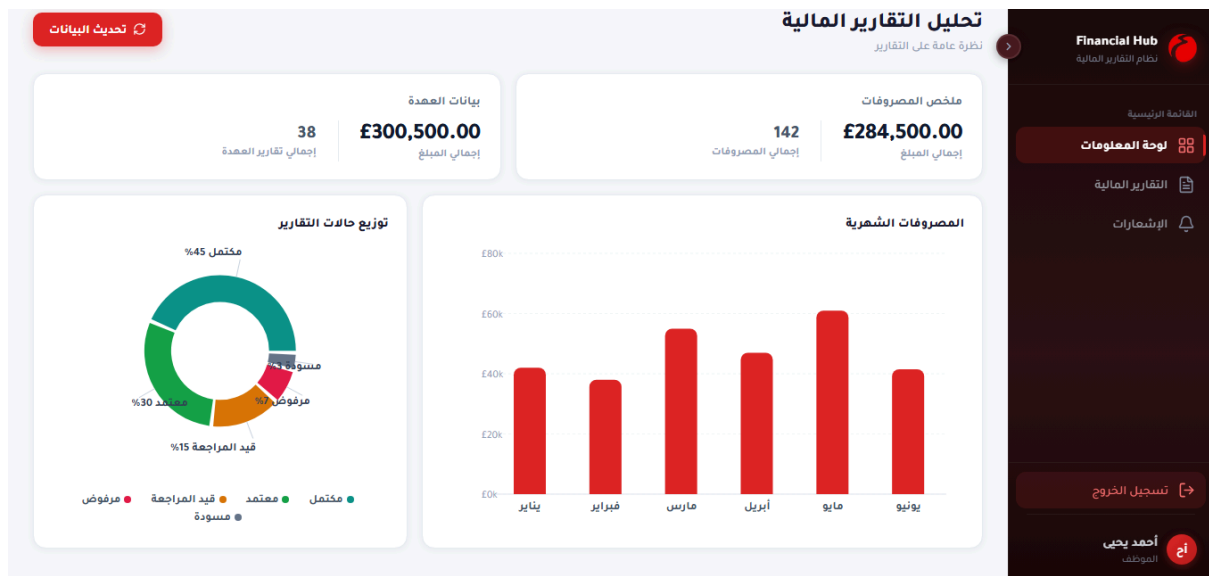
Based on the nature of the system and the features being tested, the following challenges were anticipated before testing began:

- Role-based access conflicts: Testing different user roles (Accountant, Principal, Board Member) in the same environment required careful session management to avoid permission bleed between accounts.
- Attachment handling: File upload and retrieval functionality depended on a correctly configured backend file directory. A missing or misconfigured uploads folder would cause 'Page not found' errors when accessing attachments.
- Category-specific filtering: The dashboard and records page filter records by category. With multiple categories in the system, ensuring each filter returned only the correct results was expected to require multiple test rounds.
- Dynamic page titles: The Settlement Data page title was designed to update based on the selected category. Testing this required verifying the title update across all available categories.
- In-app chat notification flow: Testing the chat feature required logging in as two different users (Accountant and Principal) and verifying that messages triggered correct notifications for the other party.

### 3.6 Linkable Prototype

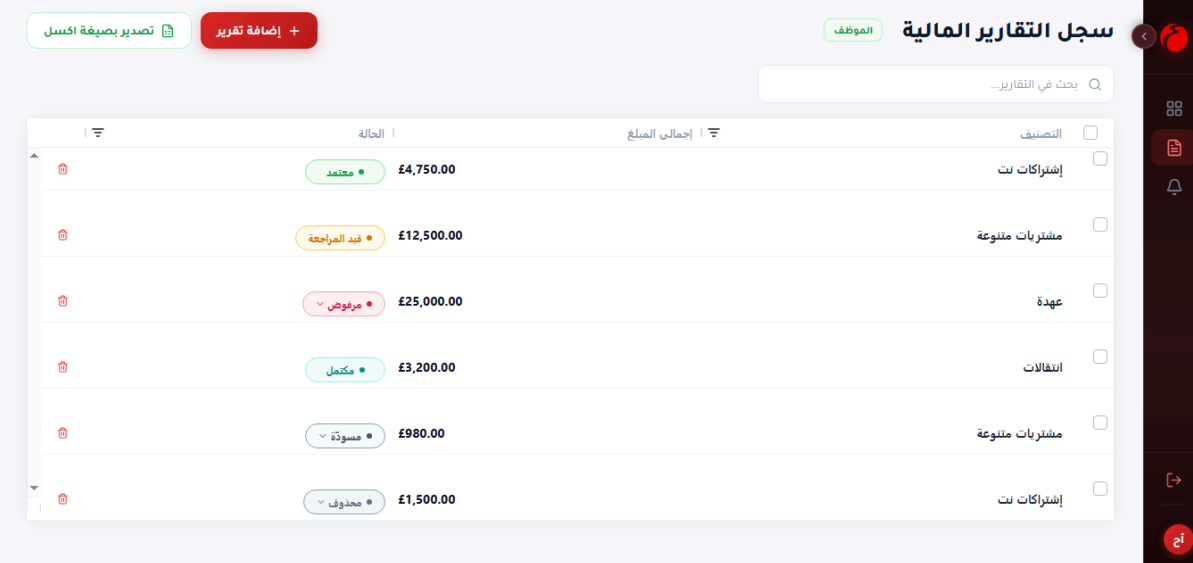
#### [Prototype Link](#)

The Financials Hub prototype is a high-fidelity, interactive web application designed to demonstrate the complete lifecycle of a school's cash disbursement process. Below are the descriptions of the three core pages that define the system's functionality and user experience.



## 1. The Unified Dashboard (Strategic Overview)

The Dashboard serves as the central command center for all users, with its content dynamically adapting to the logged-in role. For administrators and board members, it features high-level visual analytics, including a Pie Chart showing expense distribution by category and Summary Cards for total, pending, and approved records. The interface is designed for "at-a-glance" decision-making, featuring a robust search bar with date-range and category filters, allowing users to drill down into specific financial periods instantly.



The screenshot displays the 'سجل التقارير المالية' (Financial Reports Register) interface. It includes a search bar at the top right and a table of records. The table has columns for 'الحالة' (Status), 'إجمالي المبلغ' (Total Amount), and 'التصنيف' (Category). The records are as follows:

| الحالة       | إجمالي المبلغ | التصنيف        |
|--------------|---------------|----------------|
| معتد         | £4,750.00     | إشتراكات نت    |
| قيد المراجعة | £12,500.00    | مشتريات متنوعة |
| مرفوض        | £25,000.00    | عقود           |
| مكتمل        | £3,200.00     | استقالات       |
| مسودة        | £980.00       | مشتريات متنوعة |
| محذوف        | £1,500.00     | إشتراكات نت    |

## 2. Financial Records Register (The "Source of Truth")

This page acts as the primary ledger for the institution. It utilizes a structured table view where each row represents a unique Cash Disbursement Record. Users can track the real-time status of any record (e.g., Draft, Under Review, Approved, or Rejected). For the Accountant, this page is the starting point for creating new entries, while for the Principal, it serves as the review queue. The inclusion of status-based filtering ensures that no record is lost in the transition between staff and administration.

→ العودة إلى سجل التقارير

يمكنك إرفاق الملفات فقط - سيتم التوقيع التوقيع "مكتمل" تلقائياً عند الإرفاق

بيانات تسوية إشتراكات نت

أضف وصفاً مختصراً (اختياري)

إشترارات نت التصنيف

أشترارات نت

أشترارات نت بواسطة أحمد يحيى

سجل المصروفات

إرفاق الملفات فقط

| م | التاريخ    | المبلغ | البيان                     | المرفقات            |
|---|------------|--------|----------------------------|---------------------|
| 1 | 01/20/2024 | 300    | شحن نت المدرسة 01252652642 | فاتورة.pdf<br>إرفاق |
| 2 | 01/22/2024 | 0.0    | أدخل البيان...             | إرفاق               |

### 3. Settlement Data & Transaction Entry (The Operational Core)

This is the most detailed page of the prototype, where the actual financial data entry occurs. It features an integrated AG Grid component that allows accountants to enter multiple transaction rows with precision. Each row captures the date, amount, description, and a link to the digital attachment (invoice/receipt). A key "Blue-level" feature is the Dynamic Title, which updates based on the selected category (e.g., "Settlement Data: Internet Subscriptions"), and the Automatic Summary Section, which calculates totals and balances in real-time to eliminate manual calculation errors.

Building the Financials Hub prototype involved navigating complex design and technical trade-offs. Our team focused on three primary challenges that defined the system's final quality:

1. **Role-Based Logic & UI Clarity:** The most significant challenge was ensuring the interface remained simple while enforcing strict security. We had to determine exactly which buttons and data fields should be visible to an Accountant versus a Principal. We solved this by adopting an "Empathy-First" design approach—mapping out the exact needs of each user role in Figma before writing a single line of code. This ensured that the Principal is never overwhelmed by data entry fields, and the Accountant has a clear path to submission.
2. **Strategic Navigation: The "Popup vs. Full-Page" Decision:** Early testing featured a popup window for creating records. However, we realized this felt "unprofessional" and cramped for complex financial data. In a strategic pivot, we removed the popup entirely in favor of a dedicated, full-page Settlement Data flow. This decision provided the necessary space for the AG Grid transaction table and the automatic summary section, significantly improving the user's focus and data accuracy.

3. Data Visualization with Purpose: We initially struggled with which charts to include on the Dashboard. Rather than adding "eye candy," we researched which insights a School Owner actually needs. We settled on a Distribution Pie Chart for expense categories and Status Summary Cards. This ensures the Dashboard provides immediate "Actionable Intelligence" rather than just raw data.

### 3.7 Test and Evaluate:

#### Use Cases:

1. Check the Login and Sign-Up functionality
2. Check the Dashboard filtering and update functionality
3. Check the Cash Disbursement Records (Financial Report) search and status filtering
4. Check the Cash Disbursement record creation and settlement data entry
5. Check file attachment upload and access
6. Check the in-app chat and notification system

#### Test Cases:

| Test Case ID | Test Case Description       | Test Steps             | Input Data                                                        | Expected Results         | Actual Results                | Execution  |
|--------------|-----------------------------|------------------------|-------------------------------------------------------------------|--------------------------|-------------------------------|------------|
| 1            | Log In Test                 | Page not found.        | Page not found.                                                   | Page not found.          | Page not found.               | Page not f |
| 2            | Sign Up Test                | Page not found.        | Page not found.                                                   | Page not found.          | Page not found.               | Page not f |
| 3            | Dashboard Test (Search Bar) | Select search filters. | Category: "جميع التصنيفات"<br>Date from: 3/1/2026<br>To: 4/1/2026 | Show reports as filtered | Functions as expected         | Pass       |
| 4            | Dashboard Test (Search Bar) | Select search filters. | Category: "إشتراكات نت"<br>Date from: 3/1/2026<br>To: 4/1/2026    | Show reports as filtered | Does not function as expected | Not pass   |
| 5            | Dashboard Test (Search Bar) | Select search filters. | Category: "مشتريات متنوعة"<br>Date from: 3/1/2026                 | Show reports as filtered | Does not function as expected | Not pass   |

|    |                                   |                                  |                                                             |                          |                               |          |
|----|-----------------------------------|----------------------------------|-------------------------------------------------------------|--------------------------|-------------------------------|----------|
|    |                                   |                                  | To: 4/1/2026                                                |                          |                               |          |
| 6  | Dashboard Test<br>(Search Bar)    | Select search filters.           | Category: "عهدة"<br>Date from: 3/1/2026<br>To: 4/1/2026     | Show reports as filtered | Does not function as expected | Not pass |
| 7  | Dashboard Test<br>(Search Bar)    | Select search filters.           | Category: "انتقالات"<br>Date from: 3/1/2026<br>To: 4/1/2026 | Show reports as filtered | Does not function as expected | Not pass |
| 8  | Dashboard Test<br>(Search Bar)    | Select search filters.           | Category: "رواتب"<br>Date from: 3/1/2026<br>To: 4/1/2026    | Show reports as filtered | Does not function as expected | Not pass |
| 9  | Dashboard Test<br>(Search Bar)    | Select search filters.           | Category: "صيانة"<br>Date from: 3/1/2026<br>To: 4/1/2026    | Show reports as filtered | Does not function as expected | Not pass |
| 10 | Dashboard Test<br>(Reset button)  | Click on "إعادة الضبط" button    | 1.                                                          | Reset filters to default | Functions as expected         | Pass     |
| 11 | Dashboard Test<br>(Update button) | Click on "تحديث البيانات" button | .                                                           | Updates data             | Functions as expected         | Pass     |
| 12 | Cash Disbursement Test            | Select search filters.           | Category: all<br>Status: all                                | Show reports as filtered | Functions as expected         | Pass     |
| 13 | Cash Disbursement Test            | Select search filters.           | Category: "اشتراكات نت"<br>Status: all                      | Show reports as filtered | Does not function as expected | Not pass |
| 14 | Cash Disbursement Test            | Select search filters.           | Category: "مشتريات متنوعة"<br>Status: all                   | Show reports as filtered | Does not function as expected | Not pass |
| 15 | Cash Disbursement Test            | Select search filters.           | Category: "عهدة"<br>Status: all                             | Show reports as filtered | Does not function as expected | Not pass |
| 16 | Cash                              | Select search filters.           | Category: "انتقالات"                                        | Show                     | Does not                      | Not pass |

|    |                        |                               |                                         |                          |                               |          |
|----|------------------------|-------------------------------|-----------------------------------------|--------------------------|-------------------------------|----------|
|    | Disbursement Test      |                               | Status: all                             | reports as filtered      | function as expected          |          |
| 17 | Cash Disbursement Test | Select search filters.        | Category: "رواتب"<br>Status: all        | Show reports as filtered | Does not function as expected | Not pass |
| 18 | Cash Disbursement Test | Select search filters.        | Category: "صيانة"<br>Status: all        | Show reports as filtered | Does not function as expected | Not pass |
| 19 | Cash Disbursement Test | Select search filters.        | Category: all<br>Status: "مكتمل"        | Show reports as filtered | Does not function as expected | Not pass |
| 20 | Cash Disbursement Test | Select search filters.        | Category: all<br>Status: "معتمد"        | Show reports as filtered | Does not function as expected | Not pass |
| 21 | Cash Disbursement Test | Select search filters.        | Category: all<br>Status: "مرفوض"        | Show reports as filtered | Does not function as expected | Not pass |
| 22 | Cash Disbursement Test | Select search filters.        | Category: all<br>Status: "قيد المراجعة" | Show reports as filtered | Does not function as expected | Not pass |
| 23 | Cash Disbursement Test | Select search filters.        | Category: all<br>Status: "محذوف"        | Show reports as filtered | Does not function as expected | Not pass |
| 24 | Cash Disbursement Test | Select search filters.        | Category: all<br>Status: "مسودة"        | Show reports as filtered | Does not function as expected | Not pass |
| 25 | Cash Disbursement Test | Click on "إعادة الضبط" button | .                                       | Reset filters to default | Functions as expected         | Pass     |

### Testing Steps: Cash Disbursement Record

The following steps describe the exact procedure followed when testing the creation of a new Cash Disbursement record:

1. Log in to the system using the Accountant account credentials.

2. Navigate to the Cash Disbursement Records page from the sidebar.
3. Click the 'Add Cash Disbursement' button to begin creating a new record.
4. Select a category from the available options (e.g., Internet Subscriptions, Petty Cash, Transportation, or Custody).
5. In the settlement data table, enter dummy data into at least two transaction rows, including date, amount, and a brief description for each row.
6. Upload a sample attachment file to at least one transaction row and verify that the attachment icon appears correctly.
7. Review the automatically generated summary section at the bottom of the page. Verify that the 'Total Amount Paid,' 'Previous Balance,' and 'Current Balance' fields reflect the entered data accurately.
8. Save the record as a draft and confirm it appears in the Cash Disbursement Records list with the correct status badge.
9. Change the record status to 'Under Review' to submit it for principal approval.
10. Log out of the Accountant account, then log in as the Principal.
11. Navigate to the Cash Disbursement Records page and confirm the submitted record appears in the list.
12. Open the record and verify that all entered data, attachments, and the summary section are visible and accurate.
13. Send a message in the in-app comment/chat section of the record.
14. Log out of the Principal account and log back in as the Accountant.
15. Confirm that the Accountant received a notification about the Principal's message and that the message is readable within the record.

Based on testing, we refined the Dashboard to include more prominent "Status Boxes" (Pending/Approved/Rejected) and improved the AG Grid tables to allow for better filtering of financial records by date and category.

### 3.8 Survey and Evaluation: =>leave this for now

- Who is your target audience to fill the survey?
- Number of survey respondents?

#### Software application Survey:

|                                    | Very Satisfied        | Somewhat Satisfied    | Neutral               | Somewhat Unsatisfied  | Very Unsatisfied      |
|------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Application software compatibility | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Documentation                      | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Collaborate with team              | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Clarity of documentation           | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Accessibility of product support   | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Quality of product support         | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Value for money                    | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
|                                    | Very Satisfied        | Somewhat Satisfied    | Neutral               | Somewhat Unsatisfied  | Very Unsatisfied      |
| Overall reliability                | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Overall performance                | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

|                                      | Very Satisfied        | Somewhat Satisfied    | Neutral               | Somewhat Unsatisfied  | Very Unsatisfied      |
|--------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Ease of installation                 | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Ease of use                          | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Hardware compatibility               | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Operating system compatibility       | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Security                             | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Ability to integrate with other apps | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| Look and feel                        | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

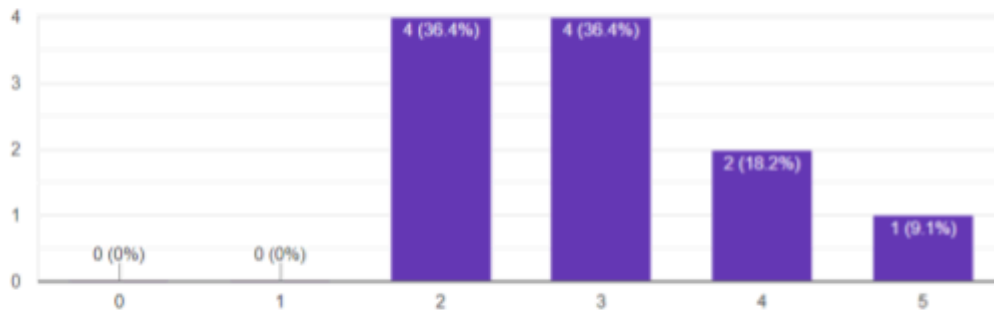
### 3.8 Refine the design. => leave this for now as I haven't made a survey yet

#### Survey Results:

3. How satisfied are you with the look and feel of this software?

[Copy](#)

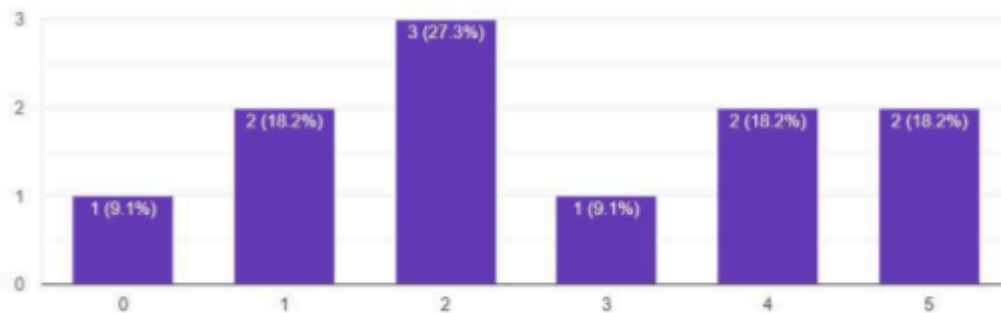
11 responses



4. How satisfied are you with the functionality of our app?

[Copy](#)

11 responses



## **Section IV: Analyzing and communicating.**

## 4.1 Recommendations

### Recommendations for Future Work

Financials Hub was designed and built as a focused prototype addressing the core financial workflow of school cash disbursements. While the system demonstrates the essential features needed for structured record creation, approval, and tracking, there are several directions in which the project could be meaningfully extended.

- **Mobile Responsiveness:** The current prototype is designed primarily for desktop use. Future development should prioritize a fully responsive layout, or a dedicated mobile application, given that school staff often work across multiple devices.
- **Advanced Dashboard Analytics:** The dashboard currently shows basic summaries and a chart. A more developed version could include trend analysis over time, comparisons between categories, monthly spending summaries, and exportable reports in PDF or Excel format.
- **Multi-School Deployment:** The system is currently scoped for a single institution. Extending it to a multi-tenant architecture would allow the same platform to serve multiple schools, each with isolated data and independent user accounts. This is the logical step toward a commercial SaaS model.
- **Audit Trail and History Log:** A detailed, tamper-proof log of every action taken on a record — who changed what and when — would significantly increase the system's value for accountability and compliance purposes.
- **Full Authentication and Security Hardening:** The current prototype uses basic authentication. A production-ready version should implement industry-standard security practices such as JWT token refresh cycles, role-permission middleware, and input validation to protect against common vulnerabilities.
- **Automated Notifications via Email or SMS:** The in-app notification system works well for users who are already logged in. Adding email or SMS alerts would ensure that principals and accountants are notified promptly even when they are not actively using the system.
- **Category and User Management Panel:** Currently, categories and user accounts are managed at the database level. A proper admin panel would

allow school administrators to add or remove categories, manage user roles, and configure the system without developer involvement.

## Advice for a Team Continuing This Project

If another team were to continue developing Financials Hub from where we stopped, we would offer the following guidance:

- Start by resolving the existing bugs identified during testing before adding any new features. The search and filter functionality on both the dashboard and the records page needs to be fixed first, as it affects the core usability of the system. Specifically, category-based filtering and case-insensitive search must be addressed.
- Fix the uploads folder configuration on the backend host so that attachments can be correctly accessed. This is a simple infrastructure fix but has a significant impact on the testing and demonstration experience.
- Do not underestimate the complexity of role-based logic. When building out new features, always test each one from the perspective of all three user roles to avoid permission-related bugs.
- Document your API endpoints clearly from the beginning. We found during testing that unclear endpoint naming caused confusion between the frontend and backend teams. A shared API documentation file (even a simple one) would have saved significant debugging time.
- Involve a real stakeholder — such as a school accountant or administrator — in at least one review session per sprint. The terminology and workflow decisions we made were greatly improved once we got feedback from someone who actually manages school finances.

## 4.2 Learning Outcomes

### Transferred Academic, Technical, and Key Competences

During the development of *Financials Hub*, our team applied knowledge and skills from multiple competencies and integrated them into a complete system.

### Competency Related

- **Wireframe & System Design:**  
We designed a simple and clear user interface using wireframes to define the layout of pages such as the dashboard and financial records.
  - **Use Case Diagram:** helped us identify system users (staff, principals, owners) and their interactions.

- **Activity Diagram:** was used to model the workflow of creating, reviewing, and approving financial records.
  - **Requirements Gathering:** allowed us to understand the real needs of school users and translate them into system features.
  - **Advanced Programming (C#) & API Development:**  
We used C# to build the backend of the system and developed APIs to handle data operations, role-based access, and communication between different parts of the system.
  - **Database Design & Management:**  
We designed and implemented a structured database to store financial records, transactions, and categories, ensuring data consistency and easy retrieval.
  - **Testing:**
    - **Test Scenarios:** were created to simulate real user actions such as creating records and approving transactions.
    - **Test Cases:** were used to verify that each feature works correctly according to the requirements.  
These steps helped us ensure the system functions reliably and meets user expectations.
- 

## Project Story

The idea of *Financials Hub* started from observing how financial processes in schools depend heavily on paperwork, spreadsheets, and manual follow-ups. This created delays, confusion, and a lack of clear tracking.

Through analysis and discussions, we realized that schools need a **simple and specialized system** that focuses specifically on financial workflows, rather than complex general systems.

This led us to develop Financials Hub, a web-based solution that organizes financial records, structures approval processes, and improves communication between different roles in the school.

---

## Learning Outcomes

Through this project, we learned how to apply theoretical knowledge in a real-world context. We improved our skills in system design, backend development, and database structuring.

We also learned the importance of:

- Understanding user needs before designing solutions
- Working collaboratively and dividing tasks effectively

- Testing and refining the system to improve performance

Additionally, we developed better time management and problem-solving skills throughout the project.

---

### Impact on Users and Market

Financials Hub improves the way schools manage their financial operations by replacing manual processes with a structured digital system. It helps staff organize records more efficiently, reduces errors, and minimizes delays in approvals.

For administrators and school leaders, the system provides better visibility through dashboards and clear tracking, supporting faster and more informed decision-making.

On a broader level, the project reflects the increasing need for **simple, focused financial management solutions in educational institutions**, contributing to digital transformation in the education sector.

## 4.3 Communication

**Abstract**

Financial management in educational institutions is often hindered by inefficient manual processes, fragmented data systems, and limited transparency. These challenges lead to delayed approvals, increased human error, and reduced operational efficiency for school administrators. This project introduces **Financials Hub**, a centralized digital platform designed to streamline financial operations through automation, real-time data integration, and role-based access control. The system was developed using prototyping tools such as Figma and supported by system design diagrams to ensure usability and scalability. Testing was conducted through structured evaluation plans and is currently ongoing, with a real-user survey phase in progress. Preliminary findings indicate that the system has strong potential to improve efficiency, reduce errors, and enhance financial transparency.

**Results and data analysis**

**Graph 1: Current Financial Management Method**  
**Type:** Pie Chart  
**Result:** 60% of respondents rely on spreadsheets, 25% use paper records, and only 15% use digital syst  
**Analysis:** This shows that most schools still depend on manual or semi-digital methods, which supports the need for a structured system like Financials Hub.

**Graph 2: Difficulty in Managing Financial Records**  
**Type:** Bar Chart  
**Result:** 70% of users reported that managing financial records is difficult or very difficult.  
**Analysis:** This indicates a clear problem in current systems and highlights the need for a more organized solution.

**Conclusions**

The Financials Hub system addresses key inefficiencies in financial management within educational institutions by replacing manual processes with automated and integrated solutions. The proposed system enhances accuracy, improves transparency, and streamlines workflows, aligning with the identified challenges and design requirements. The project is currently in the testing phase, with survey-based validation still in progress. Initial observations indicate that the system is well-positioned to meet its objectives, although further evaluation is required to confirm its effectiveness.

**Introduction**

Educational institutions struggle with financial management due to **manual and disconnected systems**, leading to inefficiency, delays, and limited transparency. **Figure 1** summarizes the most significant challenges identified in recent research. Manual processes represent the largest issue (45%), followed by data dispersal (20%) and slow approvals (15%), with additional concerns including lack of transparency and human error. These findings are supported by Saputra & Wibowo (2025) Financial Management Study.

**Figure 1: Primary Challenges in School Financial Operations**

Current approaches, such as spreadsheets, paper-based records, and isolated tools, provide basic functionality but lack integration and real-time access, making them inefficient and error-prone. **Financials Hub** is proposed as a centralized solution that automates workflows, integrates financial data, and provides real-time reporting. The system is designed to improve accuracy, increase efficiency, enhance transparency, and maintain ease of use.

**Business Model Canvas**

|                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                        |
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| <b>Key Partners</b> <ul style="list-style-type: none"> <li>School administrations</li> <li>Cloud hosting and infrastructure providers</li> </ul>                                                                                               | <b>Key Activities</b> <ul style="list-style-type: none"> <li>System design and development</li> <li>Financial workflow and approval logic design</li> <li>Continuous maintenance and feature updates</li> <li>Data security and access control management</li> </ul> | <b>Value Proposition</b> <ul style="list-style-type: none"> <li>Centralized transaction and report management for school finances</li> <li>Secure custody tracking across multiple transaction reports</li> <li>Real-time dashboard overview of financial activity and balances</li> <li>Real-time status tracking with role-based approval workflows</li> <li>Automatic notifications based on report status changes and alerts</li> </ul> | <b>Customer Relationship</b> <ul style="list-style-type: none"> <li>Self-service usage for report creation and management</li> <li>Automated workflow control based on report status and user roles</li> <li>In-app communication between staff and administrators per report</li> <li>System-generated notifications for status changes and required actions</li> <li>Technical support</li> </ul> | <b>Customer Segments</b> <ul style="list-style-type: none"> <li>School accountants</li> <li>School principals (approval and oversight role)</li> <li>School owners and board members (view-only oversight)</li> <li><b>Target Market:</b> Niche market - Educational institutions (schools)</li> </ul> |
| <b>Key Resources</b> <ul style="list-style-type: none"> <li>Software development team</li> <li>Financial workflow and approval logic</li> <li>Database server and storage systems</li> <li>Cloud hosting and deployment environment</li> </ul> |                                                                                                                                                                                                                                                                      | <b>Channels</b> <ul style="list-style-type: none"> <li>Web-based application (cloud-hosted)</li> <li>Partnerships with educational administrators</li> </ul>                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                        |
| <b>Cost Structure</b> <ul style="list-style-type: none"> <li>Cloud hosting and storage fees</li> <li>Development and testing effort</li> <li>Ongoing maintenance and updates</li> <li>Technical support and system administration</li> </ul>   |                                                                                                                                                                                                                                                                      | <b>Revenue Streams</b> <ul style="list-style-type: none"> <li>Annual subscription per school</li> <li>Optional paid customizations for large institutions (future extension)</li> </ul>                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                        |

**Prototype**

**Materials**

Figma: Prototype design  
Microsoft Word: Documentation  
Microsoft PowerPoint: Display board & Presentation  
Google Forms: Survey data collection  
draw.io: System diagrams  
Google Forms: Survey  
Visual Studio / VS Code: Development  
Git & GitHub: Version control

**Recommendation**

Based on system testing and stakeholder feedback, the following improvements are recommended to enhance performance, usability, and scalability:

- Notification System Expansion**  
Extend the current in-app notification system to include email and SMS alerts, ensuring that key stakeholders remain informed even when not actively using the platform.
- Administrative Control Panel**  
Design a user-friendly admin interface that enables administrators to manage expense categories, permissions, and user roles without requiring technical support.
- Scalability and Deployment Model**  
Transition the system toward a multi-tenant architecture, allowing multiple institutions to operate independently within the same platform. This supports scalability and future commercialization.

**Methods**

- Analyze financial management problems from research
- Define system requirements
- Design system using diagrams:
  - Use Case Diagram (user interactions)
  - Activity Diagram (system workflow)
- Develop interactive prototype
- Create testing plans for each requirement
- Conduct user evaluation (survey in progress)
- Prepared survey (ongoing) for real-user validation

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# Section V: Business Model Canvas Template

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